MASTER'S THESIS

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StepQuest – Development and Evaluation of a Mobile Game to Promote Everyday Physical Activity

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1 Introduction

1.1 Goal

The goal of this thesis is to design, implement, and evaluate *StepQuest*, a Unity-based mobile exergame that aims to encourage players to integrate regular walking into their everyday routines. The game links real-world step counts to in-game progress by using walking as the central resource for completing daily quests, unlocking content, and advancing through a short narrative structure. Instead of treating physical activity as an external task, *StepQuest* aims to embed walking directly into the core gameplay loop.

The project addresses the broader problem of insufficient physical activity by exploring how mobile game mechanics can support motivation for low-threshold, everyday movement. Walking was chosen as the target activity because it is accessible, requires no special equipment, and can be performed by a wide range of users in everyday contexts. By focusing on step-based progress, the prototype attempts to create a simple connection between real-world behavior and virtual rewards.

A central goal of *StepQuest* is to investigate how game design elements such as daily quests, progression, feedback, unlockable content, and narrative framing can be used to support sustained engagement. Exergame design research emphasizes that activity-based games should balance exercise-related goals with enjoyable and meaningful gameplay experiences [1], [2]. Research on gamified mobile applications further suggests that game elements can positively influence physical activity and engagement [3]. In addition, the project draws on the Octalysis framework [4], [5] and the Behaviour Change Wheel [6] to guide the selection of motivational mechanics and provide a theoretical rationale for their intended motivational and behavioural functions.

The thesis does not aim to prove the long-term effectiveness of *StepQuest* as a clinical health intervention. Instead, it evaluates the prototype as an exploratory, theory-informed mobile game and examines whether its design can create motivating experiences around walking. The evaluation focuses on user experience, engagement, motivation, and indications of walking-related behavior during the test period. In this way, the project aims to contribute practical design insights into how mobile exergames can be structured to make everyday physical activity more engaging.

1.2 Background

Insufficient physical activity is a persistent public health problem and remains a relevant challenge for digital health and human-computer interaction research. The World Health Organization identifies physical inactivity as a major risk factor for non-communicable diseases and recommends that adults perform at least 150 to 300 minutes of moderate-intensity aerobic physical activity per week [7], [8]. However, recent global data show that a large share of the adult population does not meet these recommendations. Strain et al. [9] report that the global age-standardised prevalence of insufficient physical activity among adults reached 31.3% in 2022, corresponding to approximately 1.8 billion adults worldwide [9].

The issue is also relevant in an Austrian context. According to Statistik Austria [10], only a minority of adults in Austria meet both the aerobic and muscle-strengthening recommendations defined by the WHO. This highlights the need for approaches that lower the barrier to physical activity and can be integrated into everyday routines. Walking is a suitable target activity in this context because it is low-threshold, does not require special equipment, and can be performed in many everyday situations. Research shows that walking interventions can improve several health-related outcomes, including aerobic fitness, blood pressure, and measures related to body composition [11]. Walking-based approaches are therefore well suited for interventions that aim to increase daily movement without requiring users to adopt a demanding sport routine. For a mobile intervention, walking also has the advantage that it can be measured through step counts, making it possible to connect real-world behavior directly to digital feedback and progress.

Mobile technologies offer a promising platform for such interventions because smartphones are already part of many users' everyday lives and can measure physical activity with built-in sensors. Smartphone-based interventions have been shown to increase physical activity [12]. However, these effects are often modest and may be more pronounced in the short term than over longer periods [13]. This limitation suggests that tracking and numeric goals alone may not be sufficient to sustain engagement. Digital physical activity interventions therefore need to address not only measurement and feedback, but also motivation and perceived meaning.

Games provide one possible way to address this challenge. By linking physical activity to goals, feedback, rewards, progression, and narrative context, mobile exergames can make movement part of an interactive experience rather than presenting it only as a health-related obligation. Previous research on gamified smartphone applications indicates that gamification can have positive effects on physical activity, but also shows that the effectiveness of such systems depends on the concrete design of the implemented game elements [3]. Similarly, large-scale examples such as *Pokémon Go* demonstrate that location-based and movement-based games can temporarily increase real-world activity, while also showing that novelty effects and declining engagement remain important challenges [14], [15].

This creates the research gap addressed by *StepQuest*. There is a need for mobile exergame prototypes that do not only count steps, but deliberately structure walking around motivational game mechanics such as daily quests, progression, unlockable content, and narrative feedback. *StepQuest* explores how these elements can be combined in a Unity-based mobile game to support regular walking and sustained engagement during everyday use. The importance of this research therefore lies in examining how a low-threshold activity such as walking can be connected to game design in a way that is motivating, technically feasible, and suitable for evaluation in a real-world prototype setting.

1.3 Research Questions and Hypotheses

The evaluation focuses on whether *StepQuest* can provide a motivating and engaging game experience around walking and whether indications of changed walking behavior can be observed during the test period.

Based on this objective, the thesis addresses the following research questions:

- **RQ1:** How does the use of *StepQuest* influence users' self-reported motivation to walk?
- **RQ2:** Which design elements of *StepQuest* are perceived as supporting or limiting engagement with the game?

These research questions reflect the exploratory nature of the project. *StepQuest* is evaluated as a theory-informed mobile exergame prototype, with a focus on user experience, motivation, engagement, and walking-related behavior during everyday use. The evaluation therefore combines user feedback with usage-related data collected during the test period.

The following hypotheses guide the evaluation:

- **H1:** Users will report feeling more motivated to go for walks after using *StepQuest*.
- **H2:** Users will perceive a meaningful connection between real-world walking and in-game progress.

The hypotheses are based on previous research showing that physical activity games can support motivation when they provide clear goals, feedback, progression, and meaningful rewards [1], [2]. Gamified mobile applications have likewise been shown to have a positive effect on physical activity [3].

2 Methodology

This chapter describes the methodological approach used to design and evaluate *StepQuest*. The thesis follows a Design Science Research approach in which a functional mobile game prototype serves as the research artifact. The methodology combines theory-informed game design with an exploratory user study to investigate the motivational design of the prototype and participants' experiences during real-world use.

The chapter is structured into three main parts. First, the research method is introduced and the application of the Design Science Research Methodology to *StepQuest* is explained. Second, the design rationale is presented. This includes the requirements of a walking-based exergame, a behavior-change analysis using the Behaviour Change Wheel and COM-B model, the selection and application of Octalysis for motivational game design, and the relationship between both frameworks. Third, the user study is described, including the testing procedure, data collection, and data analysis methods used to evaluate the prototype.

2.1 Research Method Selection

The aim of this thesis is to design, implement, and evaluate a functional software artifact. The methodological structure of this thesis follows the Design Science Research Methodology described by Peffers et al. [16]. This process consists of six steps: problem identification and motivation, definition of objectives, design and development, demonstration, evaluation, and communication. Their application within this thesis is summarized in Table 1.

Prototyping was chosen because the central concept of *StepQuest* cannot be fully examined on a theoretical level alone. The project combines step tracking, quest progression, unlockable content, narrative feedback, and mobile interaction. These elements need to be implemented in an interactive system before their feasibility and user experience can be meaningfully assessed. In design science research, the creation and evaluation of an artifact is considered a way to generate knowledge about a problem and a possible solution [16], [17]. For *StepQuest*, the prototype serves as this artifact: it translates the theoretical design assumptions into a concrete mobile game that can be tested by users.

The prototype also makes it possible to examine technical and practical constraints that would not be visible in a purely conceptual design. These include the reliability of step tracking on Android devices, the clarity of the quest flow, the usability of the interface, and the suitability of the daily quest structure for real-world use. Since *StepQuest* is intended to be used outside a laboratory setting, the prototype needed to be functional enough for participants to install and

Table 1: Application of the Design Science Research process to StepQuest

Design Science Step	**Application in StepQuest**
Problem identification and motivation	Insufficient physical activity and the difficulty of sustaining motivation for regular walking.
Definition of objectives	Design of a mobile exergame that connects real-world walking to in-game progress and motivational game mechanics.
Design and development	Implementation of StepQuest as a Unity-based Android prototype.
Demonstration	Use of the prototype by participants in a real-world setting during the testing period.
Evaluation	Assessment through questionnaires, usage data, and qualitative feedback.
Communication	Presentation and interpretation of the findings in the results and discussion chapters of the thesis.

use it during their normal routines.

User testing was selected as the second main method because the success of *StepQuest* depends strongly on user perception and interaction. The prototype aims to motivate walking through game mechanics such as daily quests, progress feedback, unlockable content, and narrative framing. Whether these mechanics are understandable, motivating, or frustrating cannot be determined only from the implemented features. It requires feedback from users who interact with the prototype. User testing is therefore appropriate for identifying usability issues, evaluating perceived engagement, and collecting qualitative feedback on the strengths and limitations of the design [18], [19].

2.2 Design Rationale and Framework Selection

Before implementation, the design of *StepQuest* was informed by exergame research, gamification concepts, and behavior change theory. To motivate the selection of the design frameworks used in this thesis, the following subsection summarizes the main requirements that are relevant for a walking-based exergame. These requirements describe what the design of *StepQuest* should support and provide a basis for explaining why Octalysis and the Behaviour Change Wheel were considered suitable frameworks.

2.2.1 Design Requirements for StepQuest

A central requirement was that physical activity and gameplay should form a coherent experience. Exergame research emphasizes that health-related objectives should be integrated with engaging game design rather than treated as a separate component [2]. For *StepQuest*, this meant that walking should directly contribute to meaningful in-game progress instead of functioning only as an externally imposed activity requirement.

The game also needed to provide clear goals, meaningful feedback, and visible progression, as these are important principles in the design of activity-based games [1], [2]. Players should therefore be able to understand what is required of them, observe their progress toward a walking goal, and receive a recognizable in-game outcome after completing it.

Another requirement was to minimize interaction with the smartphone during physical activity. Exergame mechanics should support the intended activity rather than interfere with it [1]. Consequently, *StepQuest* was designed around interactions before and after a walk, while step progress could be accumulated without requiring continuous attention to the screen.

The prototype was further intended to support repeated engagement over multiple days. Instead of encouraging players to complete all available content in a single session, the game needed a progression structure that provided reasons to return regularly. Daily quests, unlockable content, and gradual narrative progression were therefore considered suitable mechanisms for structuring continued use.

Finally, the design needed to consider not only whether the game itself was engaging, but also whether its mechanics could plausibly support the target behavior of regular walking. This required consideration of both the motivational qualities of the game and the factors that influence a person's capability, opportunity, and motivation to perform the desired behavior.

Based on these considerations, the main design requirements of *StepQuest* were defined as follows:

- walking should be directly connected to meaningful in-game progress;
- goals and progress should be clearly communicated to the player;
- successful walking activity should produce visible feedback and rewards;
- interaction with the smartphone during walking should be minimized;
- the progression structure should support repeated engagement across multiple days; and
- the selected mechanics should be justifiable from both a game-design and a behavior-change perspective.

2.2.2 Behaviour Change Wheel

The Behaviour Change Wheel (BCW) was used to examine the target behavior of regular walking from a behavior-change perspective. At the center of the BCW is the COM-B model, according to which a behavior occurs through the interaction of *Capability*, *Opportunity*, and *Motivation*. [6] Rather than assigning individual game mechanics directly to BCW intervention functions, the target behavior was first considered in terms of these three components.

Target Behaviour and COM-B Analysis

The target behavior of *StepQuest* is regular walking in everyday life. A conceptual COM-B analysis was used to identify factors that are relevant to performing this behavior.

Capability concerns whether a person is physically and psychologically able to perform the behavior. Walking was deliberately selected as a low-threshold form of physical activity that requires no special equipment or technical skill. Psychological capability is supported by keeping the interaction model simple: players receive a clearly defined step goal and continuous feedback about their progress.

Opportunity concerns external conditions that make the behavior possible. Physical opportunity depends on whether users have sufficient time and an environment in which walking is possible. *StepQuest* cannot directly change these environmental conditions, but the mobile format allows quests to be completed in different locations and incorporated into everyday routines. Social opportunity plays only a minor role in the prototype because no social or multiplayer mechanics were implemented.

Motivation includes both reflective processes, such as goals and intentions, and automatic processes, such as emotional responses and reward-driven behavior. This component is particularly relevant for *StepQuest*, since the central purpose of the game is to provide additional reasons to walk. Reflective motivation is addressed through explicit walking goals, visible progress, and the larger objective of recovering the ship. Automatic motivation is supported through immediate feedback, rewards, unlockable content, and anticipation of further progression.

Derived Intervention Functions

Based on the COM-B analysis, the intervention functions most relevant to *StepQuest* were *Persuasion*, *Incentivisation*, *Enablement*, and, to a smaller extent, *Education* [6].

Persuasion was considered relevant because the game attempts to create positive associations with walking through narrative framing and meaningful in-game objectives. *Incentivisation* was selected because walking results in visible rewards, including quest completion, unlockable ship parts, and story progression. *Enablement* is reflected in the low-threshold structure of the game, which allows players to complete walking goals during normal daily activities without requiring a specific location or additional equipment. *Education* is represented to a limited extent through short health-related information presented within the prototype.

Other BCW intervention functions, such as *Training*, *Modelling*, and *Coercion*, were not central to the final design. The target behavior does not require learning a complex physical skill, social modelling was outside the scope of the single-player prototype, and punitive mechanisms were deliberately avoided.

2.2.3 Octalysis

While the Behaviour Change Wheel provides a behavior-change perspective on the target behavior of regular walking, Octalysis is used to structure the motivational game design of *StepQuest* [4], [5]. The framework provides a vocabulary for relating different sources of motivation to concrete game mechanics. In the context of *StepQuest*, this is particularly useful because the intended behavior should not only be feasible, but should also be embedded in an engaging game experience.

Motivational Core Drives

Octalysis organizes motivation into eight core drives, each representing a different reason why a person may engage with a system [4], [5]. *Epic Meaning and Calling* refers to motivation that arises from contributing to a larger purpose. *Development and Accomplishment* concerns progress, mastery, and overcoming challenges. *Empowerment of Creativity and Feedback* describes motivation through meaningful choices, experimentation, and feedback on one's actions. *Ownership and Possession* is based on the desire to collect, improve, or control something of personal value.

Social Influence and Relatedness includes social comparison, cooperation, recognition, and a sense of connection to others. *Scarcity and Impatience* concerns opportunities or rewards that are limited or not immediately available. *Unpredictability and Curiosity* describes the motivation to discover what will happen next when outcomes are uncertain. Finally, *Loss and Avoidance* refers to motivation created by the desire to prevent the loss of progress, rewards, or opportunities.

The eight core drives were used as a design space for considering how walking could be connected to motivational game mechanics. This did not require every core drive to be represented equally. Instead, mechanics were considered according to their suitability for the intended gameplay loop, technical scope, and compatibility with a walking-based mobile game.

Mapping to Game Mechanics

Epic Meaning and Calling was addressed through the narrative setting of *StepQuest*. The player is stranded on an unfamiliar planet and must recover missing ship parts and information. Daily walking challenges were therefore intended to contribute to a larger in-game objective rather than functioning only as isolated step targets. The narrative gives the player's physical activity an additional purpose within the game world.

Development and Accomplishment was addressed through clearly defined step goals, progress feedback, quest completion, and visible long-term progression. Individual quests provide short-term objectives, while recovering ship parts and advancing the narrative provide a longer-term sense of progress.

Empowerment of Creativity and Feedback was considered through the possibility of allowing players to choose between walking goals with different step requirements. This provides a limited degree of autonomy by allowing players to select an available task according to their current situation. Immediate step feedback was intended to make the relationship between real-world movement and the game state clearly visible.

Ownership and Possession was addressed through collectible ship parts and story modules. Unlockable content was intended to remain permanently available in the player's collection and thereby provide a persistent representation of previous accomplishments. The gradually reconstructed ship additionally serves as a visual representation of accumulated progress.

Possible mechanics related to *Social Influence and Relatedness* included achievement sharing, trophy displays, and cooperative walking challenges. Such mechanics could introduce social recognition, comparison, or cooperation. However, they were considered secondary to the central single-player gameplay loop.

Scarcity and Impatience was considered through a limitation on daily progression. Restricting the number of quests that can be completed within one day was intended to distribute progression across multiple days and create a reason to return instead of allowing all available content to be completed in a single session.

Possible mechanics related to *Unpredictability and Curiosity* included random discoveries, mystery rewards, and short questions related to health or the game's narrative. Such elements were considered as a way of introducing variation and uncertainty into otherwise predictable walking tasks.

Finally, *Loss and Avoidance* was considered through time-limited opportunities, such as rewards or quests that would expire after a defined period. These mechanics were considered cautiously because excessive time pressure or punishment could conflict with the intended positive and low-threshold experience of *StepQuest*.

The mapping represented a set of intended design possibilities rather than a requirement to implement every mechanic. The final selection was influenced by feasibility, technical complexity, safety considerations, and compatibility with the intended walking experience.

2.2.4 Combined Design Rationale

The Behaviour Change Wheel and Octalysis address different but complementary aspects of the design of *StepQuest*. The BCW focuses on the target behavior and provides a structured way to consider the capability, opportunity, and motivation required for regular walking. Based on these factors, relevant intervention functions can be identified. Octalysis, in contrast, focuses on the motivational qualities of the game experience and provides a more detailed vocabulary for translating motivational intentions into concrete game mechanics.

Several relationships between the two perspectives can be observed in the design of *StepQuest*. The BCW intervention function *Incentivisation*, for example, is reflected in Octalysis through *Development and Accomplishment* and *Ownership and Possession*, which are represented through quest completion, progress feedback, and collectible content. *Persuasion* is supported by *Epic Meaning and Calling*, as the narrative gives walking a meaningful in-game purpose. Elements of *Enablement* are supported through clear goals, feedback, and a limited degree of quest choice, while *Education* can be incorporated through health-related information.

Other combinations were considered but played a smaller role in the prototype. Social mechanics associated with *Modelling* could have supported *Social Influence and Relatedness*, while time-limited opportunities associated with *Coercion* could have contributed to *Loss and Avoidance*. These approaches were considered less suitable for the primary scope of the prototype.

2.3 User Study

A user study was conducted to investigate participants' experiences with *StepQuest* during everyday use. The study focused on perceived walking motivation, engagement with the prototype, usability, and the perceived connection between real-world walking and in-game progression. Questionnaire data were complemented by usage logs and metrics recorded by the prototype.

The study followed an exploratory field-study design without a control group. It was therefore not intended to establish a causal effect of *StepQuest* on physical activity. Instead, the study examined how participants perceived the prototype, which elements contributed to motivation and engagement, and how the application was used during the testing period.

2.3.1 Participants and Testing Procedure

Study materials were distributed to 14 individuals. Participation was restricted to adults and required an Android smartphone on which the *StepQuest* APK could be installed. Participation was voluntary, and informed consent was collected through the pre-questionnaire before study data were collected.

Distribution began on June 11, 2026. Participants received an email containing the Android APK together with a link to the pre-questionnaire. After installing the prototype, participants were asked to use *StepQuest* in their everyday environment rather than under controlled laboratory conditions. The overall data-collection period ended on July 31, 2026.

After their individual testing period, participants were asked to complete a post-questionnaire. In addition, a dedicated export function was implemented in *StepQuest* that allowed participants to send the locally recorded application logs and metrics to the researcher by email. The attached log files were downloaded from these emails and subsequently processed without retaining identifying email information for the analysis.

The questionnaires were collected anonymously. Consequently, pre- and post-questionnaire responses could not be matched at an individual level. Questionnaire results were therefore treated as separate descriptive samples rather than as paired measurements.

2.3.2 Data Collection

Data were collected from three sources: a pre-questionnaire, a post-questionnaire, and usage data recorded by the prototype.

The pre-questionnaire collected demographic information including age, gender, occupation, and health-related limitations affecting walking. Baseline physical activity was assessed using the short form of the International Physical Activity Questionnaire (IPAQ-SF) [20]. The IPAQ-SF assesses physical activity performed during the previous seven days and distinguishes between vigorous activity, moderate activity, walking, and sedentary time. It was chosen because it provides a standardized and comparatively short instrument for characterizing participants' physical activity while keeping the burden of the pre-questionnaire low. The instrument has been evaluated across multiple countries and has shown acceptable test-retest reliability for population-level physical activity monitoring [20]. For the analysis of walking behavior in this study, particular attention was given to the reported number of days per week on which participants walked for at least ten minutes at a time. As the IPAQ-SF is based on self-reported activity and has limited agreement with objective activity measurements [21], it was used to characterize baseline activity rather than to obtain a precise objective measure of individual physical activity.

The pre-questionnaire additionally collected information about participants' frequency of playing digital games and previous experience with games or gamified applications for health, fitness, or learning. Five-point Likert-scale items ranging from *Strongly disagree* to *Strongly agree* assessed initial walking motivation, perceived ability to integrate walking into everyday routines, confidence in completing walking goals, and attitudes toward using game elements to support walking. An open-text question asked participants what they expected would make a walking game motivating.

The post-questionnaire used the same five-point Likert response format to assess perceived walking motivation, whether walking felt more meaningful or rewarding through *StepQuest*, and whether the daily quests encouraged additional walking. Further items assessed engagement with the narrative and progression system, clarity of the relationship between steps and game progress, usability, step-progress feedback, technical friction, willingness to use a more complete version, and willingness to recommend the concept. Participants also selected the game element they perceived as most motivating and provided open-text feedback about positive aspects, frustrations, and desired improvements.

In addition to questionnaire data, *StepQuest* recorded usage-related information locally during gameplay. At the end of the testing period, participants could export the recorded logs and summary metrics directly from the application by email. These data were used to describe actual interaction with the prototype and complement the participants' self-reported experiences.

2.3.3 Data Analysis

Because of the exploratory study design and the small sample size, the quantitative data were analyzed descriptively. Questionnaire responses were summarized using absolute frequencies and percentages. For individual Likert-scale items, response distributions and medians were used rather than relying solely on arithmetic means.

Pre- and post-questionnaire responses were not analyzed as paired observations because the questionnaires were anonymous and did not contain a shared participant identifier. Accordingly, no statistical claim of individual pre-to-post change was made. Baseline variables from the pre-questionnaire were instead used to characterize the study sample, while the post-questionnaire was used to assess participants' perceived influence of *StepQuest* on motivation and engagement.

The exported usage logs were analyzed descriptively to identify patterns of interaction and continued use during the testing period. The recorded metrics were aggregated across anonymized log files and used to complement the questionnaire findings.

Open-text questionnaire responses were reviewed and grouped according to recurring topics. The purpose of this qualitative analysis was to identify common positive aspects, sources of frustration, technical problems, and suggestions for future development rather than to conduct a formal large-scale thematic analysis.

3 Results

3.1 Prototype Outcome

The development process resulted in a functional Android prototype that implements the central *StepQuest* gameplay loop. Players can select a walking quest, accumulate progress through real-world steps, complete the quest after reaching the required step target, and unlock ship parts or story-related content. The prototype also preserves progress between sessions, restricts players to one completed quest per day, and records usage-related information for the user study via logging.

3.1.1 Technical Overview

StepQuest was implemented using Unity version 6000.2.10f1 with the Universal Render Pipeline (URP) and C#. Unity was used to implement the game logic, user interface, asset management, local persistence, and Android build process. The final application was distributed as an Android Package Kit (APK), allowing participants to install the prototype directly on compatible Android smartphones without requiring publication through an application store.

The application was configured with Android API level 29 as the minimum supported version, while the target API level was set to Unity's *Automatic (highest installed)* option. The minimum API level was selected to provide broad compatibility with currently used Android devices while maintaining support for the APIs required by the prototype.

Android was selected as the primary target platform because it provides access to built-in motion sensors for step counting and supports direct distribution of APK files for prototype testing. *StepQuest* accesses the Android sensor system to obtain step-related activity data and integrates these data into the Unity-based quest and progression systems.

Table 2 summarizes the main technologies used in the development of the prototype.

3.1.2 Software Architecture

The prototype was divided into separate systems for step tracking, quest management, persistence, user interface control, content unlocking, and data logging. This separation reduced dependencies between the physical activity sensor and the gameplay logic. For example, the quest system receives step updates through a common step-counter interface rather than directly accessing the Android sensor implementation.

Table 2: Main technologies used in the *StepQuest* prototype

Technology	**Purpose**
Git	Version Control
Unity	Game engine, interface implementation, scene and asset management, and Android builds
C#	Implementation of the gameplay logic, quest system, persistence, and logging
Android Sensor API	Access to step counter and step detector sensors
Local device storage	Persistence of active quests, unlocked content, and daily progress
APK distribution	Direct installation of the prototype on participants' Android devices

The step-counter interface has separate implementations for Android devices and the Unity editor. On Android, sensor data are obtained from the device's native step sensors. During development in the Unity editor, a mock implementation allows steps to be added manually. This made it possible to test the complete quest flow without performing physical walks during each development iteration.

The central quest state is managed by a dedicated quest manager. This component stores the currently active quest, the number of accumulated steps, and the unlocked content. It also notifies interface components when a quest is selected, progress changes, or a quest is completed.

Quest content is represented through Unity ScriptableObjects. Each quest definition contains configuration data such as a unique identifier, step requirement, displayed text, quest type, and corresponding unlockable content. Separating this information from the gameplay code made it possible to add and modify quests without changing the underlying quest logic.

3.1.3 Step Tracking

Real-world walking progress is measured using the motion sensors available on Android devices. The prototype first attempts to use the Android `TYPE_STEP_COUNTER` sensor. This sensor reports the cumulative number of steps recorded since the device was last restarted [22]. When a quest or tracking session begins, *StepQuest* stores the current cumulative value as a baseline. Subsequent progress is calculated as the difference between the current sensor value and this baseline.

If the cumulative step counter is unavailable, the prototype can use the `TYPE_STEP_DETECTOR` sensor as a fallback. In this case, individual step events are counted within the current session [23]. Both sensor types have been available since Android API level 19. On Android 10 (API level 29) and later, access to these sensors requires the `ACTIVITY_RECOGNITION` permission [24].

The measured steps are forwarded to the quest system and used to update the active quest. The user interface displays the current number of steps relative to the quest target. Once the required value is reached, the quest becomes eligible for completion.

Because sensor availability and behavior can differ between Android devices, step tracking represented an important technical constraint of the prototype. The implementation was designed to support the two common Android step sensor types, but reliable tracking still depended on the hardware and operating-system behavior of the participants' devices.

3.1.4 Quest and Progression System

The central gameplay loop is implemented through a daily quest system. Each quest defines a step target and an associated in-game reward. During the introductory phase, the repair-droid quest is mandatory because it establishes the narrative context and introduces the relationship between real-world walking and in-game progression. After this initial quest, the player can select from the available quests.

The implemented step targets range from approximately 1,000 to 5,000 steps. These values were chosen as a design compromise between accessibility and perceived effort rather than as prescribed physical-activity targets. Shorter quests were intended to remain achievable within everyday routines, while higher targets provide greater variation and a more substantial challenge. The objective was therefore not to prescribe a specific amount of daily physical activity, but to provide walking goals that could be completed within a realistic period while still requiring deliberate activity.

The prototype distinguishes between two main quest types: ship-part quests and story quests. This distinction was introduced to provide two complementary forms of progression. Ship-part quests provide a tangible and visually persistent reward by unlocking components of the damaged spacecraft. In contrast, story quests unlock information modules that extend the narrative and reveal additional context about the game world. The two quest types therefore combine collection-based progression with narrative progression, reducing reliance on a single type of reward and addressing different motivational aspects of the game.

Quest progress is communicated through a numerical step display and a progress bar. When the required number of steps is reached, the quest can be completed and its associated content is added to the player's collection. Collected ship parts remain visible and gradually reconstruct the spacecraft, providing a persistent representation of previous accomplishments and long-term progress.

The prototype restricts progression to one completed quest per calendar day. This limitation was introduced to distribute the available content across multiple days rather than allowing the complete prototype to be finished in a single session. Since repeated engagement was one of the intended characteristics of the design, the daily limit creates a recurring gameplay rhythm in which players select, complete, and receive the reward for one walking objective before returning on a later day. The restriction was kept simple and predictable so that missing a day does not result in the loss of previously earned progress. Technically, the date of the most recently completed quest is stored locally and compared with the current calendar date. Once a quest has been completed, another quest cannot be completed until the following day.

3.1.5 Persistence and Data Logging

The prototype stores the game state locally on the Android device. Persisted data include the active quest, current quest progress, unlocked content, the date of the most recently completed quest, and the starting date of the playthrough. This allows users to close and reopen the application without losing their progress.

A separate logging system was implemented for the user study. The recorded data describe interaction with the prototype and include quest-related events, step progress, quest completion, unlocked content, and relevant application state changes. In addition, summary metrics derived from these events were made available for export. The application itself did not require a user account and did not store names, email addresses, or other direct personal identifiers as part of the game state or log data.

Both the persistent game state and the study logs were stored locally on the participant's device. No data were automatically transmitted to a remote backend or research server. Instead, participants could initiate an export using a dedicated function within the application. This function created the log and metric files and opened a dialog for sending the exported data. After receipt, the attachments were downloaded and processed separately from identifying email information.

Because the export was mainly transmitted by email, the sender's email address could initially be associated with the received files. After downloading the attachments, identifying email information was not retained as part of the analysis dataset. The log and metric files were subsequently processed without names, email addresses, or participant account information and were treated independently of the sender information during analysis.

Local storage and participant-initiated export were chosen because the prototype did not require user accounts, synchronization between devices, or a persistent backend infrastructure. This reduced the amount of participant-related information collected by the application and avoided continuous transmission of usage data. The trade-off was that study data had to be exported individually from each participant's device after the testing period.

3.1.6 User Interface

The final prototype uses a portrait-oriented mobile interface divided into three primary areas: the home screen, the narrative modules screen, and the ship collection screen. Navigation is provided through a tab-based bar at the bottom of the display.

The home screen presents the current quest, step requirement, and progress. It also contains the dialogue and narrative elements used before and after quest completion. The collection screen shows recovered ship parts, where the user can interact with his ship by rotating it to view from all sides. The narrative modules screen shows the unlocked modules, which provide further explanation of the worlds setting via a text based pop up. Locked content is visually distinguished from unlocked content so that players can see both their current progress and the remaining rewards.

Interaction during walking was intentionally minimized. Players can select a quest before beginning a walk and then carry the smartphone in a pocket while steps are recorded. The interface is mainly used before and after the physical activity rather than requiring continuous attention during movement.

The resulting prototype therefore implements the main functional requirements needed for the user study: physical step tracking, daily walking goals, progress feedback, quest completion, content collection, narrative progression, persistence, and usage logging.

3.1.7 Mapping of Implemented Game Mechanics to Octalysis and the BCW

The final *StepQuest* prototype implemented a subset of the design ideas discussed during the design phase. To distinguish the roles of Octalysis and the Behaviour Change Wheel, Table 3 maps the central implemented game mechanics independently to both frameworks. Octalysis is used to describe the motivational role of a mechanic, whereas the BCW is used to describe its intended behavior-change function. Not every mechanic requires a direct mapping to both frameworks.

The mapping shows that the two frameworks describe different aspects of the same design. For example, unlocking a ship part can be interpreted through *Development and Accomplishment* and *Ownership and Possession* in Octalysis because it provides both progression and a persistent collectible reward. From a BCW perspective, the same mechanic represents *Incentivisation*, since completion of the walking task produces an in-game reward. Similarly, quest selection and progress feedback relate to *Empowerment of Creativity and Feedback* in Octalysis, while their behavioral role is more closely associated with *Enablement*.

Some implemented mechanics do not have a meaningful direct equivalent in both frameworks. The one-quest-per-day restriction, for example, is primarily a pacing and scarcity mechanic and was therefore mapped to *Scarcity and Impatience*, without assigning it an additional BCW intervention function. Conversely, the short health-related information corresponds to the BCW function *Education* but was not a central Octalysis mechanic. This distinction avoids forcing every design element into both frameworks.

Several mechanics considered during the design phase were not implemented in the final prototype. Social features such as leaderboards, achievement sharing, and cooperative quests were excluded because they would have required additional account, networking, privacy, and

Table 3: Mapping of implemented *StepQuest* mechanics to Octalysis and BCW

Game Mechanic	**Octalysis Core Drive**	**BCW Intervention Function**
Narrative framing and overall objective	*Epic Meaning and Calling*	*Persuasion*
Step goals and visible quest progress	*Development and Accomplishment*; *Empowerment of Creativity and Feedback*	*Enablement*
Quest completion and unlockable rewards	*Development and Accomplishment*	*Incentivisation*
Collectible ship parts and persistent collection	*Ownership and Possession*	*Incentivisation*
Choice between available quests	*Empowerment of Creativity and Feedback*	*Enablement*
Gradual story progression and information reveals	*Epic Meaning and Calling*; limited *Unpredictability and Curiosity*	*Persuasion*
One-quest-per-day progression limit	*Scarcity and Impatience*	–
Short health-related information	–	*Education*

moderation systems and did not fit the intended isolated narrative setting. Consequently, *Social Influence and Relatedness* and the BCW function *Modelling* were not substantially represented.

Random rewards, mystery boxes, and similar mechanics related to *Unpredictability and Curiosity* were also not implemented. Curiosity was instead addressed only indirectly through gradual narrative reveals. Finally, the prototype did not implement loss-based mechanics such as expiring rewards, removal of progress, or punishment for missed days. *Loss and Avoidance* and the BCW intervention function *Coercion* therefore did not form part of the final motivational structure.

Overall, the implemented design focused on narrative purpose, visible progress, rewards, collection, limited player choice, and daily progression pacing. Octalysis describes how these mechanics contribute to the motivational game experience, while the BCW provides a separate perspective on how selected mechanics may support the target behavior of regular walking.

3.2 User Study Results

3.2.1 Participation and Sample Characteristics

A total of 14 pre-questionnaire responses were initiated. Twelve complete responses with documented consent were included in the analysis. One completed response was excluded because consent to participate was not provided, and one additional response was incomplete. Seven participants completed the post-questionnaire.

The twelve participants included in the pre-questionnaire analysis had a median age of 26 years, with ages ranging from 22 to 57 years. Eleven participants identified as male and one as female. Seven participants were employed full-time, four were students, and one was employed part-time. None reported a health-related limitation affecting walking.

At baseline, participants reported walking for at least ten minutes on a median of four days per week. The interquartile range was approximately four to five days, with responses ranging from zero to seven days per week.

The sample also had substantial previous exposure to digital games. Six participants reported playing digital games daily or almost daily and three reported playing three to five times per week. Six participants had not previously used a game or gamified application for health, fitness, or learning.

Initial attitudes toward walking and gamification were generally positive. Eight of twelve participants agreed or strongly agreed that they were motivated to walk more in their daily lives. Nine considered walking realistic to integrate into their everyday routine. Nine agreed that using a game to motivate walking sounded interesting, while ten expected story progression and rewards to make walking more engaging.

3.2.2 Questionnaire Results

Seven complete post-questionnaires were available for analysis. Table 4 summarizes the response distributions for the closed Likert-scale items.

Responses concerning walking motivation were mixed. Three of seven participants agreed that they felt more motivated to walk after using *StepQuest*, while four disagreed. Four participants agreed that the game made walking feel more meaningful or rewarding, while the remaining three selected the neutral response. Three participants agreed that the daily quests encouraged them to go outside or walk more than usual, two were neutral, and two disagreed.

Responses concerning engagement were more consistently positive. Six of seven participants agreed or strongly agreed that they wanted to continue the story and recover additional parts. All seven participants agreed that the connection between real-world steps and game progress was clear.

Usability was also rated positively. All seven participants agreed or strongly agreed that the application was easy to understand and that they knew what to do next. Six participants rated

Table 4: Post-questionnaire response distributions ($n = 7$)

Item	Disagree	Neutral	Agree	Median
More motivated to walk	4	0	3	2
Walking felt more meaningful or rewarding	0	3	4	4
Daily quests encouraged more walking	2	2	3	3
Wanted to continue the story and recover parts	1	0	6	5
Connection between steps and game progress was clear	0	0	7	5
Application was easy to understand	0	0	7	5
Always knew what to do next	0	0	7	5
Step-progress feedback was clear and useful	1	0	6	5
Technical issues made the application harder to use	3	1	3	3
Would use a more complete version again	1	0	6	5
Would recommend the concept	0	0	7	5

the step-progress feedback as clear and useful. Experiences with technical problems were less consistent: three participants agreed that technical issues made the application harder to use, three disagreed, and one selected the neutral response.

Six of seven participants indicated that they would use a more complete version of *StepQuest* again. All seven agreed or strongly agreed that they would recommend the concept to someone who wanted to walk more.

When asked which element of *StepQuest* was most motivating, three participants selected story progression and three selected unlocking ship parts or the collection system. One participant selected the daily walking goal. No participant selected the step-progress bar or "Nothing in particular" as the primary motivator.

3.2.3 Usage and Engagement Metrics

3.2.4 Qualitative Feedback

The open-text responses largely supported the positive ratings of narrative progression and collectible content. Participants frequently identified the story, collecting ship parts, and observing visible progress as the aspects they liked most. Immediate rewards for everyday movement were also mentioned positively.

The most frequently reported negative experiences concerned technical issues with step tracking and progress registration. Participants described delayed or unreliable step updates and, particularly during earlier versions of the prototype, problems with progress when the application was not kept open. Feedback also identified the restriction to one quest per day as a source of frustration for some participants.

Suggestions for future development included improving the reliability of the step counter, increasing the amount and difficulty of available content, adding additional game systems such as leaderboards, and providing more flexibility in daily progression. One participant suggested linking walking distance to exploration of an in-game map. Several responses therefore indicated interest in expanding the prototype beyond the relatively limited progression system implemented for the study.

4 Discussion

4.1 Interpretation of the Main Findings

4.2 Walking Motivation and Engagement

4.3 Research Questions and Hypotheses

4.4 Limitations and Future Work

4.4.1 Limitations

4.4.2 Future Work

A Software Artifacts

A.1 Source Code Repository

The source code of the StepQuest prototype is available in the following GitHub repository:

https://github.com/straussdave/StepQuest

Release version 0.3.5 was used during the user study

A.2 Android Application Package

The APK used during the user study is available at:

https://drive.google.com/file/d/1Pk4RrTVzy1FXJfj_MVrzTgyCyz_CZW8W/view

The APK can be installed on compatible Android devices by downloading the file and permitting installation from the application used to open it. The prototype requires access to physical activity data for step counting.

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Documentation table of AI-based tools

AI-Based Tool	Intended Use	Prompt, Source, Page, Paragraph...
Chat-GPT 5.x	Image Generation	"make a portrait of a non humanoid, simple repair droid with orange and blue accents", Prototype
Chat-GPT 5.x	Technical Implementation	General assistance in the implementation of the prototype in Unity, Prototype
Chat-GPT 5.x	Language	"Improve academic phrasing, grammar, structure, and clarity of provided text.", Entire document

List of Abbreviations

API	Application Programming Interface
APK	Android Package Kit
BCW	Behaviour Change Wheel
COM-B	Capability, Opportunity, Motivation-Behaviour
H	Hypothesis
RQ	Research Question
WHO	World Health Organization